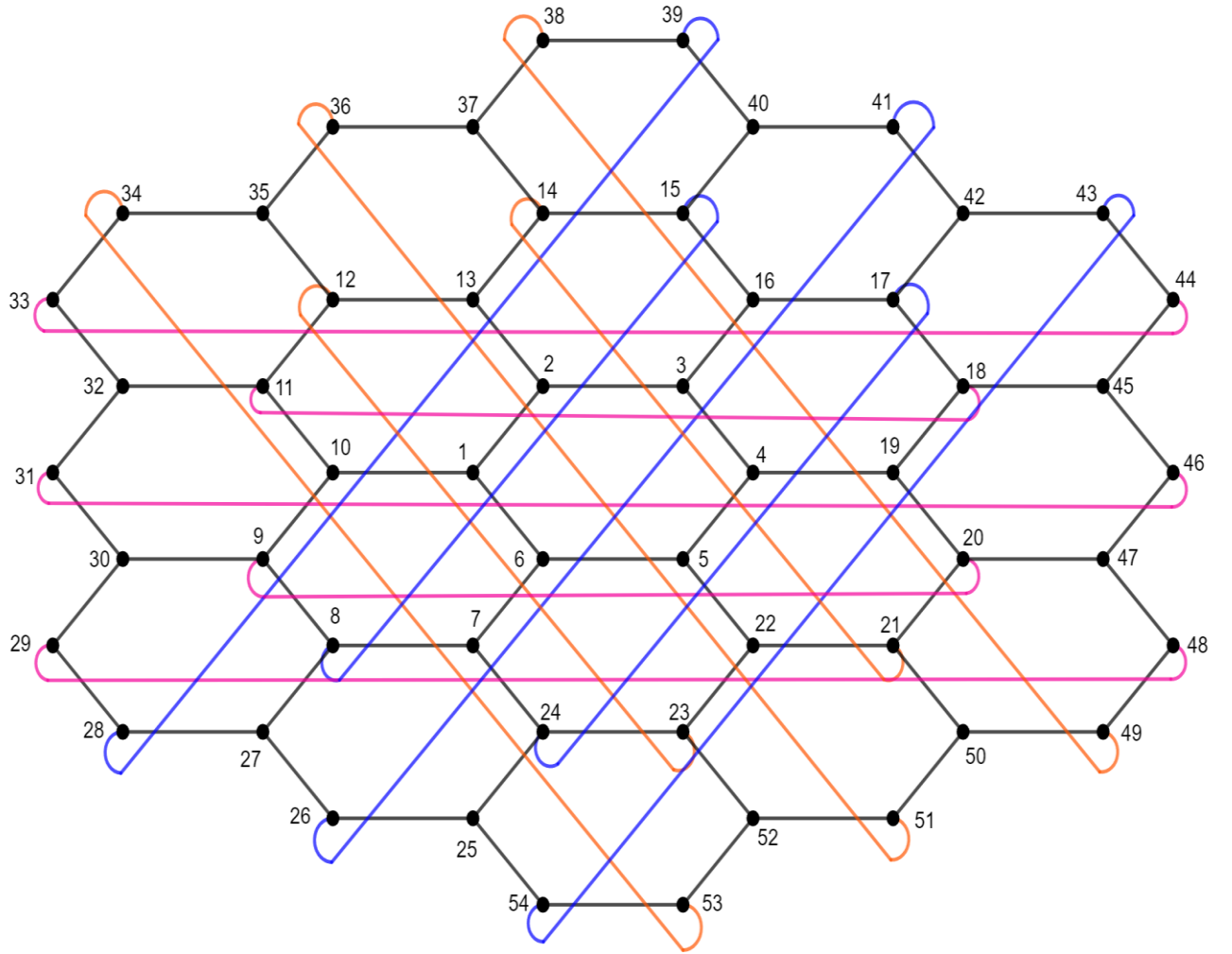


Extended Honeycomb Torus network of dimension three $EHCT(3)$



1-fault tolerant metric dimension - 6

$IDRW$ index - 9

Extracted 1-fault tolerant metric bases using extended $BIGS$ algorithm

- | | |
|---------------------------|-----------------------------|
| 1. {1, 3, 5, 28, 38, 48} | 6. {2, 5, 25, 32, 40, 47} |
| 2. {1, 3, 5, 34, 44, 54} | 7. {3, 6, 30, 37, 45, 52} |
| 3. {1, 4, 27, 35, 42, 50} | 8. {28, 33, 38, 43, 48, 53} |
| 4. {2, 4, 6, 29, 39, 49} | 9. {29, 34, 39, 44, 49, 54} |
| 5. {2, 4, 6, 33, 43, 53} | |